Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building platform automation, CI/CD, and containerized services in **Python** on **Linux**, **Docker**, and **Kubernetes**, plus autonomous-driving ML pipelines over 100K+ trajectory logs and production LLM workflow orchestration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **LLM workflow** orchestration with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under load.
- Operated recurring **data pipelines** across PostgreSQL, SQL Server, and cloud storage with validation and reconciliation checks, running 5K+ monthly jobs at 99%+ production reliability through CI/CD-backed releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Ran a production Python platform on AWS RDS serving 10K+ daily requests, adding timeout-aware failover across external model providers and per-call usage and **cost tracking** for every model invocation.
- Built concurrent batch processing with ThreadPoolExecutor over 1K+ files per run with fault-tolerant execution, shipping through GitHub Actions **CI/CD** and cutting feedback turnaround 60% for 150+ instructors.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built end-to-end **autonomous-driving** data and training pipelines processing 100K+ trajectory logs and HD-map vectors for a GPT-based offline-RL Decision Transformer, improving long-horizon trajectory planning by 18% in closed-loop simulation.
- Ran reproducible PyTorch experiments with Hydra configs and W&B tracking and benchmarked encoders in closed-loop **simulation** pipelines, improving closed-loop planning stability by 15% while working within tight single-GPU compute limits.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated nightly **Jenkins** pipelines that allocated VMs, pulled commits, built and installed executables, and ran tests on Linux, Windows, and macOS, emailing failure logs and coverage reports.
- Ran parallel builds on India and US servers with branch-based workflows in GitHub, and redesigned AWS Glue and S3 ETL, boosting pipeline efficiency 68% and cutting **storage costs** 15%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Deployed eight Docker-packaged microservices to **Kubernetes** on infrastructure provisioned with Ansible, shipping through Jenkins CI/CD with manual approval gates and rolling zero-downtime updates, reaching 95%+ build validation across releases.
- Ran platform **observability** with Sentry for errors, Azure Monitor for metrics, and Kibana for logs across eight production services synchronizing 2M+ records across multiple locations over real-time WebSockets.

Skills

Languages: Python, TypeScript, JavaScript, Java, Bash, Shell, SQL, C++
Platform & Infrastructure: Linux, Docker, Kubernetes, Ansible, Terraform, ARM Templates, NGINX, TCP/IP, Git
CI/CD & Workflows: Jenkins, GitHub Actions, CI/CD, Async Job Orchestration, ETL Pipelines, Databricks, Spark
Cloud & Data: AWS (EC2, S3, RDS, ECS/Fargate, Glue), Microsoft Azure, GCP, PostgreSQL, MongoDB
Observability: Sentry, Azure Monitor, Kibana, Grafana, Logging, Monitoring
AI & ML: LLM Orchestration, PyTorch, Simulation Pipelines, Hydra, Weights & Biases

Projects

Simulation Job Platform on Kubernetes | *Python, Kubernetes, Argo Workflows, Terraform, MQTT, Prometheus*

- Designed a **Kubernetes** platform where Argo Workflows fan out simulation and ML jobs with autoscaled node pools provisioned by **Terraform**, MQTT job events, and Prometheus and Grafana cost dashboards.

Linguistix | *Python, Docker, AWS ECS/Fargate, CodePipeline, CI/CD*

- Containerized with Docker and deployed a Python REST data processing API on **AWS** ECS/Fargate with automated CodePipeline CI/CD releases and validation checks, achieving 87% accuracy and production-grade reliability.

FinNexus Live | *Kubernetes, Next.js, TypeScript, MongoDB, WebSockets*

- Managed **Kubernetes** deployments and CI/CD automation for a TypeScript and Next.js fintech platform with real-time WebSocket updates, serving 50K+ users across 1M+ transactions with a token-based firewall.